High-frequency Ventilation

Manuel Sánchez-Luna, MD, PhD[a,*],
Noelia González-Pacheco, MD, PhD[a], Martín Santos-González, DVM, PhD[b],
Francisco Tendillo-Cortijo, DVM, PhD[b]

KEYWORDS

• High-frequency ventilation • Volume guarantee • Ventilation efficacy
• CO_2 elimination • Tidal volume

KEY POINTS

- High-frequency ventilation (HFV) is an alternative to conventional mechanical ventilation (CMV) with less risk of ventilator-induced lung injury (VILI) and more effectivity in washout CO_2 from the lungs.
- Rescue high-frequency oscillatory ventilation (HFOV) has been used when severe respiratory distress syndrome needs aggressive CMV settings to prevent lung damage.
- HFOV is more effective to open atelectatic lungs but also has a role in air leak syndrome, meconium aspiration syndrome, and lung hypoplasia.
- Today it is possible to measure and to set directly high-frequency tidal volume (VThf), which can make it possible to decrease VThf to protect the immature lung from large tidal volumes and to decrease fluctuations in the VThf.
- This new strategy can be used in preterm infants with RDS using very low VThf and high frequencies to maintain ventilation and decrease the risk of VILI.

INTRODUCTION

Very immature infants are at risk of lung damage soon after delivery due to exposure to oxygen and distension of the lung by breathing air. Knowledge that ventilator-induced lung injury (VILI) in premature newborn infants with respiratory failure can trigger bronchopulmonary dysplasia (BPD) recently led to modifications in the respiratory support applied to this population for a gentler and less invasive mechanical ventilation, and the combination of initial lung stabilization with early surfactant therapy has been shown to decrease the most severe forms of BPD.[1]

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[a] Neonatology Division, Instituto de Investigación Sanitaria Hospital General Universitario Gregorio Marañón, Complutense University of Madrid, Madrid, Spain; [b] Medical and Surgical Research Unit, Instituto de Investigación Sanitaria Puerta de Hierro–Segovia de Arana, Madrid, Spain
* Corresponding Author. Neonatology Department, C/ Dr. Esquerdo 46, E-28007 Madrid, Spain.
E-mail address: msluna@salud.madrid.org

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But in some severe cases, using invasive mechanical ventilation still is needed. Most of the new ventilators can synchronize with the inspiratory effort using sophisticated trigger systems. Conventional mechanical ventilation (CMV), due to the need to use large tidal volume (TV) and high ventilator pressures, can damage the immature lung. High-frequency ventilation (HFV) emerged as an alternative to conventional ventilation in the 1970s, because this technique can be beneficial due to its efficacy in recruiting a collapsed lung and wash out more CO_2 with less lung trauma. Although HFV can reduce the incidence of VILI and BPD, the evidence of the potential benefits of its elective use is low, due to use of different ventilation protocols and devices and clinical situations,[2] so it has been considered a rescue therapy. Decades later, the use of a high lung-volume strategy and the possibility of measuring and controlling, in a very precise manner, high-frequency TV (VThf) during HFV with new ventilators[3] offer a new alternative to CMV, mostly in the more immature infants, to decrease the risk of lung damage during invasive ventilation.[4]

THE HISTORY OF HIGH-FREQUENCY VENTILATION

HFV is defined as the use of a constant distension pressure to maintain a high-lung volume, normally over the functional residual capacity while transmitting energy to the airway by an oscillatory pressure amplitude (ΔP) around the mean airway pressure (Paw) to produce TVs smaller than the dead space (VThf) delivered at supraphysiologic frequencies (fr).[5] This is possible when the lung is inflated to near its total lung capacity; then, the Paw can be reduced maintaining the lung volume and applying oscillatory pressure at the maximal curvature of the deflation limb of the pressure-volume relationship of the lung. This area is considered the safe area for ventilation.[6]

Although different types of HFV devices are in use, most experimental and clinical studies in neonates use high-frequency oscillatory ventilation (HFOV). Other devices, such as high-frequency flow interrupters (HFFIs) and high-frequency jet ventilators (HFJVs), are used less frequently in neonates and less information is available.

The HFJV delivers short pulses of pressurized gas directly into the upper airway through a narrow-bore cannula or jet injector, and exhalation is a result of passive lung recoil.[7]

With this modality, because exhalation is passive, a theoretical increase in the risk of gas trapping is possible, but no evidence of more gas trapping exists when comparing HFJV and HFOV in premature infants.[8]

HFFI ventilators create, over a continuous distending pressure, bursts of gas not directly into the airway but into the ventilator circuit at some distance back from the trachea and endotracheal tube, and exhalation also is passive.

HFOV works at frequencies between 3 Hz and 20 Hz (180–1200 cycles/min). Over a Paw, the ventilator produces small TV (VThf), smaller than the anatomic dead space, 2.4 mL/kg (range 1.0–3.6), that reach the airway by a piston or a membrane that create oscillatory pressure around the Paw; thus, the gas goes in and out to and from the airway actively, with an active exhalation unlike HFJV and HFFI.[9]

Explanations by which HFOV maintains normal gas exchange have been offered extensively, with multiple involved mechanisms to explain how the use of TVs smaller than the conducting airways volume can induce gas exchange. These mechanisms include convection (dependent on bulk flow), diffusion (spontaneous gas mixing due to brownian motion), and the combined effects of both mechanisms.[10]

In some instances, the VThf can gain alveolar areas and contribute to ventilation in a conventional way, with direct penetration of fresh gas into the alveolar space might being responsible in part for the improved gas exchange efficiency.[11]

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HFOV is more effective than conventional ventilation for CO_2 removal,[12] because this is related to the square of VThf,[13,14] generated by variations on ΔP[15] and transmitted distally into the airways. This transmission is easier when the airway is short and wide without reduction in the airway diameter. In adults, due to increased lung size and bronchial branching, DP is not well transmitted, ΔP is not well transmitted and the efficiency of the HFOV decreases,[16] making the HFOV more effective and probably less injurious in neonates compared with in adults.

The use of a high-volume strategy decreases the risk of neurologic lesions, described initially in some studies where this strategy was not used.

In the HIFI study, where no high-volume strategy was used, HFOV was associated with an increased incidence of pneumoperitoneum of pulmonary origin, grades 3 and 4 intracranial hemorrhage, and periventricular leukomalacia compared with CMV.[17]

In a meta-analysis of elective use of HFOV in preterm infants compared with CMV, the short-term neurologic morbidity with HFOV was found only in the subgroup of 2 trials not using a high-volume strategy with HFOV.[2]

ELECTIVE USE OF HIGH-FREQUENCY VENTILATION

As evidenced in immature animal models,[18–20] small TVs and lung volume maintenance are able to prevent VILI in the preterm lung, and HFOV has been proposed as an ideal mode of ventilation to prevent damage of the immature lung; but, more than 40 years after the initial use of HFOV, there still is no consensus about the potential benefits of using electively HFOV in immature infants to prevent lung damage.

Earlier studies did not report measurement of VThf because most of the devices were not able to monitor TV or at all.

Although in some clinical studies, VThf as high as 3 mL/kg has been reported in neonates, exceeding the anatomic dead space and permitting some direct alveolar ventilation,[21] use of VThf of more than 2 mL/kg also is described, with these high TVs a possible explanation of why HFOV still put the infant lung at risk of damage during ventilation.[22]

More often, the frequency used and recommended by clinicians is 10 Hz, but, if there is CO_2 retention, then traditionally the frequency is reduced to less than 10 Hz to send a larger VThf,[23] because TV decreases when higher frequency is used.[24] A multicenter randomized controlled trial in 2002[25] demonstrated a higher rate of successfully extubation at 30 days, alive and weaned from all respiratory support at 36 weeks' postmenstrual age and less pulmonary hemorrhage in the HFOV group compared with the CMV group. In this study in premature infants of less than 1200 g, the ventilator device (SensorMedics) as in the original article only appears SensorMedics was a SensorMedics 3100A ventilator for HFOV, with an inspiratory to expiratory (I:E) ratio of 1:3, a frequency of 10 Hz to 15 Hz, and a lung recruitment strategy for an ideal lung inflation as expansion to 8 to 9.5 ribs on the chest radiography and compared with CMV using synchronized flow ventilation with expiratory TVs of 4 mL/kg to 7 mL/kg. No differences in other outcomes were found, demonstrating the safety of the HFOV when the lung recruitment strategy was used.

At the same time, a UK trial[26] randomized patients of less than 29 weeks' gestation to HFOV or CMV as the primary mode of ventilator support. In this study, different ventilators in different centers were used in the HFOV group with optimized lung volume, the Dräger Babylog 8000 (Dräger, Lübeck, Germany), the SensorMedics 3100A, or the SLE 2000 HFO (Specialised Laboratory Equipment Ltd, South Croydon, UK), with an I:E ratio of 1:1 in the Dräger and the SLE ventilators and 1:3 with the SensorMedics, and an initial frequency of 10 Hz modifying the ΔP to adjust $Paco_2$. No differences

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in the composite primary outcome of death or chronic lung disease (defined by a dependence on supplemental oxygen at 36 weeks' postmenstrual age) were found between the 2 groups. The effect of the type of ventilators was analyzed, and statistically differences in the frequency of the primary outcome between infants treated with the SensorMedics ventilator and those treated with either of the other 2 models were found, with a better result in the patients where these other ventilators.

These investigators followed the studied population, who at 11 years to 14 years of age demonstrated a superior lung function in the group who had undergone HFOV, compared with those who had received conventional ventilation, with no evidence of poorer functional outcomes.[27]

Some observational studies using a high-volume strategy described a higher survival rate at a mean of 2.6 years of follow-up in the HFOV group of premature infants, with a mean gestational age of 27 weeks, compared with CMV, but with no information about the HFOV settings.[28]

HIGH-FREQUENCY OSCILLATORY VENTILATION AS A RESCUE THERAPY

Because the potential benefit of HFOV is to prevent lung damage related to volutrauma, RDS of premature infants is the most frequent indication. Although meta-analyses comparing rescue HFOV or HFJV therapy demonstrated no long-term benefits over CMV,[29–31] due to its efficacy in decreasing CO_2 and generating lung recruitment, there are other conditions that can be benefit from HFOV. The most frequent are[32]

- Air leak syndrome (ALS)
- Meconium aspiration syndrome (MAS)
- Lung hypoplasia (LH)
- Pulmonary hypertension of the newborn (PPHN)

LUNG HYPOPLASIA

Congenital diaphragmatic hernia (CDH) represents a clinical situation where lung hypoplasia can be the limiting survival factor. The larger randomized controlled trial comparing the effect of HFOV as elective therapy to CMV failed to demonstrate any benefit and the trial was stopped early after an interim analysis.[33] In this international multicenter study, CDHs diagnosed prenatally were treated electively using CMV or HFOV; no statistically significant difference in the combined outcome of mortality or BPD between the 2 ventilation groups were found, and secondary outcomes favor the use of CMV. Probably, the use of HFOV in these conditions has to be individualized and apply when low TVs are preferred or CO_2 removal is a limiting factor.

THERAPY FOR AIR-LEAK SYNDROME

Controlled trials demonstrated that the use of a short inspiratory time decreases the incidence of pulmonary interstitial emphysema, and, in animal neonatal models, the use of higher frequencies on HFOV decreases more the leakage than lower frequencies.[34]

Some mechanisms are described for the potential benefits of using HFOV in the ALS, as a decrease in the driving pressure of the gas through the site of the leak, or to decrease of the site size due to the short inspiratory time.

In some cases, HFOV can prevent the use of a thoracic tube to drain the air leak with a positive evolution.[35]

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The HiFO study, a multicenter, prospective, no crossover, randomized trial to determine whether HFOV would decrease the development or progression of ALS in infants with severe RDS, using a frequency of 15 Hz, with a 1:2 I:E ratio and no background tidal breaths, demonstrated a significant lower risk to develop ALS but no effect in decreasing the worsening of the air leak when it was previously established.[36]

In the meta-analysis of elective HFOV in preterm infants,[2] however, ALS occurred more frequently in the HFOV group, but most of the studies used frequencies less than 15 Hz and no data about the I:E ratio were described.

NEONATAL MECONIUM ASPIRATION SYNDROME

Although meconium staining of amniotic fluid occurs in up to 15% of all deliveries, meconium aspiration syndrome, described as respiratory distress in an infant born through meconium-stained amniotic fluid, appears in only 1.5% to 8% of those cases[37] but can be a life-threatening disorder in newborn infants. Coexistence of asphyxia and pulmonary hypertension are considered more important than the obstruction of the airways and/or damage to the lung produced by meconium. Although some experimental data suggest that HFOV can be superior to CMV,[38] and sometimes is recommended as a rescue therapy in severe MAS,[39] there are no clinical data demonstrating its potential benefit over CMV, so currently it is not recommended routinely.[40]

PPHN is a syndrome characterized by severe hypoxemia due to right-to-left shunting of desaturated blood through the foramen oval and/or the patent ductus arteriosus, not responding to supplemental oxygen. In some cases, this is due to parenchymal lung disease and recruitment of the lung with HFOV can improve its outcome.[41]

Also, the combined therapy of HFOV, surfactant, and inhaled nitric oxide can decrease the need of ECMO in the more severe forms. A better vehiculation of the inhaled nitric oxide by the HFOV can be achieved and a better respond sometimes is obtained, but cases need to be individualized.[42]

VOLUME GUARANTEE COMBINED WITH HIGH-FREQUENCY OSCILLATORY VENTILATION—A NEW ERA

VThf is proportional to the ΔP generated in each cycle and the length of the inspiratory time and is inversely proportional to the oscillation frequency; thus, traditionally, an increase in ΔP or a decrease in frequency to finally increase VThf has been used to improve CO_2 clearance.[11,15] Also, the frequency has an important role in CO_2 removal[43] and has an independent effect on the distribution of the gas within the airways[44] and to changes in the volume when oscillating at or near the resonant frequency of the respiratory system.[45]

Measurement of the VThf gives an important advantage,[16] because there is a close correlation of the VThf and the CO_2 washout,[46] described as the diffusion coefficient of CO_2 (DCO_2) which is related to the $VThf^2$ and the frequency, as follows:

$$DCO_2 = VThf^2 \times fr$$[15,47]

Traditionally, HFOV devices did not measured the VThf, so looking at the transmission of the oscillation to the thorax of the patient has been used to clinically control VThf. Today it is possible not only to measure but also to control and fix the VThf (volume guarantee [VG]) to maintain it as constant, similar to the VG described for conventional ventilation (**Fig. 1**), enabling an independent adjustment of the VThf and the

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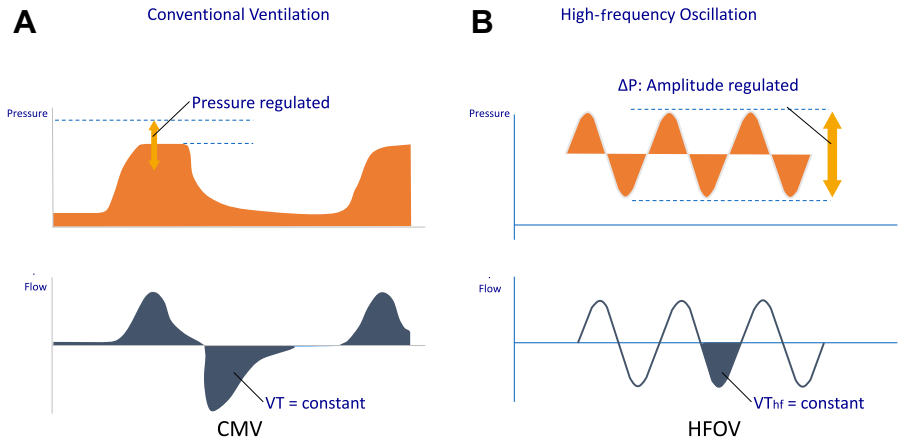


Fig. 1. Schematic representation of the VG modality, from the Dräger Babylog VN500. (*A*) Pressure regulation to control the TV set in CMV with the VG modality. (*B*) ΔP regulation to control the VThf set in HFOV with the VG modality.

frequency.[3] The ventilator, using this new technology, modifies ΔP to maintain the VThf at the setting value (**Fig. 2**).

When the VThf is fixed, any change in the frequency does not affect the VThf generated (**Fig. 3**) but directly modifies DCO_2 and CO_2 washout in the same direction as in conventional ventilation.[48]

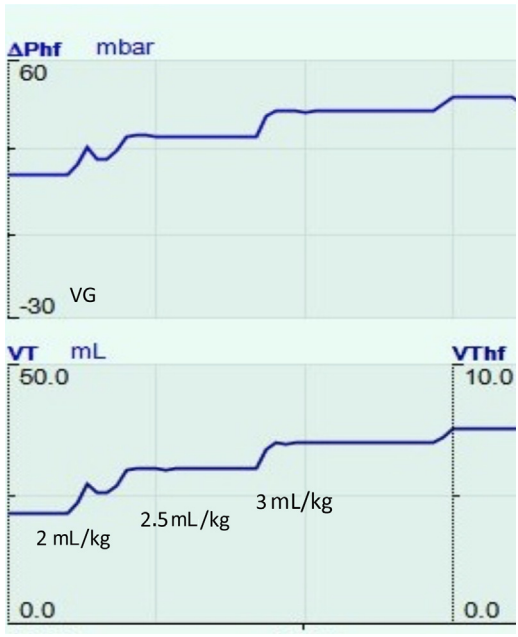


Fig. 2. Effect of increase in the VThf set on the ΔP of the ventilator during HFOV. Animal neonatal model with intact lung. Graphic obtained from the software of the Dräger Babylog VN500.

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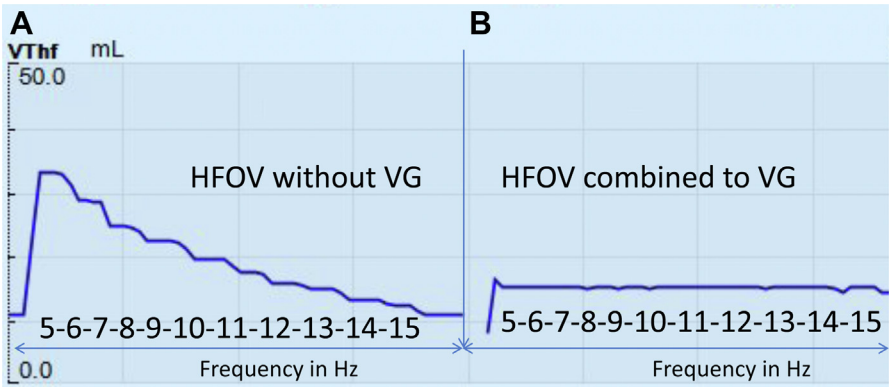


Fig. 3. (*A*) Effect of increase in the frequency during HFOV without VG. In this case, there is a drop in the VThf generated by the ventilator. (*B*) Effect of increase in the frequency during HFOV with VG. There is no change in the VThf generated by the ventilator because it is fixed. Graphic obtained from the software of the Dräger Babylog VN500, attached to a test lung.

To maintain a constant VThf, the ventilator increases the proximal ΔP when the frequency is increased, but this increase in the proximal ΔP is not transmitted into the distal airway,[16] so the lung is protected from ΔP, mostly because at lower frequencies pressure oscillations are "passed" from the airway opening to the alveolar spaces, while high-frequency pressure oscillations are heavily attenuated by resistive losses.[16,49] Also, the I:E ratio affects in a different manner if the VG is in use or not. When the HFOV does not use the VG modality, the longer the inspiratory time, the higher the VThf generated (I:E at 1:1), so more CO_2 washout is achieved at any frequency. But during HFOV combined to VG, a shorter inspiratory time (I:E at 1:2) is more effective to decrease CO_2 because the VThf is fixed and the only limitation is not having enough time to deliver the VThf set at very high frequencies.[50]

Some new generation devices can work with the VG modality, but differences in performance are described[51] as the capability of compensating the air leak around the endotracheal tube, making the VThf measured inaccurate,[52] so, it is important to use only devices that work with this leak compensation.[53]

CLINICAL RESULTS WITH HIGH-FREQUENCY OSCILLATORY VENTILATION COMBINED WITH VOLUME GUARANTEE

Because it is possible to decrease VThf in HFOV and fix it with the VG modality, DCO_2 can be maintained as constant by increasing the frequency of the ventilator as the VThf is decreased. This effect recently was demonstrated in a neonatal animal model of RDS, where the use of an approach of very low VThf at high frequencies produced a lung protective effect, demonstrated by a lower histologic damage score.[54] This new strategy has been demonstrated to be feasible in newborn infants with respiratory failure, even extremely immature infants.[55] In very immature infants, it was possible to increase the standard fr from 10 Hz to near 20 Hz maintaining similar DCO_2 and P_{CO_2} with a very low VThf without side effects.

In a randomized crossover clinical trial, Iscan and colleagues[56] found lower fluctuations of the VThf and the P_{CO_2} , similar findings by Belteki and colleagues[57] in a single-center study.

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Tuzun and colleagues,[58] in a retrospective study of preterm infants of less than 32 weeks' gestation with respiratory failure, studied the ventilator settings of HFOV combined with VG and showed a mean optimal airway pressure after lung recruitment of 10.2 mbar ± 1.7 mbar and a mean VThf of 1.61 mL/kg ± 0.25 mL/kg to provide normocapnia with no significant correlation between P_{CO_2} levels and VThf (per kilogram). VThf levels to maintain normocarbia were significantly lower with 12-Hz frequency compared with 10-Hz frequency (1.50 mL/kg ± 0.24 mL/kg vs 1.65 mL/kg ± 0.25 mL/kg; $P<.001$, respectively). The authors demonstrated this effect in a neonatal animal model, because the efficacy of CO_2 removal clearly increases with increments in the frequency more than expected, so the correlation between the P_{CO_2} and DCO_2 decreases if using the standard formula, $DCO_2 = VThf^2 \times fr$.

At frequencies higher than 15 Hz, similar DCO_2 measurements produce higher CO_2 washout, demonstrating a more effective alveolar ventilation; so, the correlation between DCO_2 and P_{CO_2} throughout increasing frequencies was not linear, showing a greater CO_2 elimination efficiency at higher frequencies, despite maintaining a constant DCO_2. To prevent this, the authors proposed a corrected formula of DCO_2 ($cDCO_2$) to be used at higher frequencies, which implies a better correlation of DCO_2 and P_{CO_2}:

$$cDCO_2 = VThf^{1.78} \times fr^{1.15}$$ [59]

This is probably due to modification of the gas flow behavior.[16] It has been suggested that the VThf during high-frequency oscillation steadily increases with increasing frequencies, reaching the maximum at the resonant frequency and subsequently decreasing again once that frequency is exceeded.[60]

The potential long-term benefits of this new modality remain to be demonstrated but there are some data showing this possibility.

Chen and colleagues[61] used HFOV combined with VG in 18 preterm infants as a rescue therapy and found a decrease in the combined outcome of BPD and death.

The authors compared a baseline period when HFOV was used without VG (2012–2013) against a period in which this strategy had been fully implemented (2016–2017). A total of 182 patients were exposed to invasive mechanical ventilation in the first 3 days after delivery, a higher proportion on HFOV at day 3 in the second period 79.5% (35) in 2016 to 2017 versus 55.4% (n 31) in 2012 to 2013. After adjusting for perinatal risk factors, the second period was associated with an increased rate of survival free of BPD (odds ratio [OR] 2.28; CI 95%, 1.072–4.878); this effect was more evident in neonates born at a gestational age of less than 29 weeks (OR 4.87; 95% CI, 1.9–12.48).[62]

A proposal of early rescue conventional ventilator settings to use HFOV combined to VG in premature infants is shown in **Table 1**, with the following initial setting of the HFOV combined with VG of the ventilator, always with a high-volume lung strategy (**Table 2**).

The lung high-volume strategy achieved using a lung recruitment protocol also is validated by other investigators.[56,63]

It is summarized as follows:

- Initial setting of the Paw 1 mbar to 2 mbar higher than that received during CMV (or similar for infants with ALS) and same fraction of inspired oxygen (Fio_2)
- Subsequently, slight increases in Paw (1–2 mbar every 2 min) until the critical lung opening pressure is reached, defined as an increase in oxygen saturation without signs of cardiocirculatory compromise
- Fio_2 then is decreased below 0.4 to 0.6, maintaining an oxygen transcutaneous saturation (Spo_2) of 90% to 95%.

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Table 1
Proposed clinical criteria for switch to high-frequency oscillatory ventilation from conventional mechanical ventilation as a rescue therapy in severe respiratory failure in newborn infants

Severe respiratory insufficiency refractory to CMV[a]: Requirements of Fio_2 >50%[b] and/or $Paco_2$ >55–60 mm Hg that needs peak inspiratory pressure of	• >15 mbar in <1000 g • >15–17 mbar between 1000 g and 1500 g • >17 mbar between 1500 g and 2.000 g • >20 mbar between 2000 g and 3000 g • >25–30 mbar in >3000 g
ALS (interstitial emphysema, pneumothorax, or pneumomediastinum)	
PPHN wherein CMV fails	
Pulmonary atelectasis	
Pulmonary hypoplasia with CO_2 retention	

[a] In preterm infants with RDS, failure of CMV is defined only after administering an initial dose of surfactant.
[b] In term infants, it is necessary to individualize and evaluate the cause of hypoxemia.

- After decreasing Fio_2, Paw is decreased by 1 mbar to 2 mbar every 2 min until Spo_2 less than 90%, defining the critical lung closing pressure.
- Paw then is increased again, to reopen the lung to the known critical lung opening pressure, and then decreased at 2 mbar above the critical lung closing pressure.
- Ideal lung inflation is defined as expansion to 8 ribs to 9 ribs (the top of the right hemidiaphragm relative to the posterior ribs on chest radiography at full inspiration).

Recently, the authors described a new clinical technique to effectively recruit the lung.

Table 2
Recommended initial settings of high-frequency oscillatory ventilation combined with volume guarantee

Fio_2	The same as in CMV
Paw	1–2 mbar > Paw in CMV Usual initial Paw for working in a safe zone: <1.000 g: 8–10 mbar 1000–1500 g: 10–12 mbar 1500–2000 g: 12–15 mbar >2000 g: >15 mbar[a]
Frequency	8–20 Hz: <2000 g: 14–20 Hz >2000 g: 8–15 Hz
ΔP	40–50 mbar
VThf	1.5–2 mL/kg
Inspiratory time	I:E 1:2 (33%) I:E 1:1 (50%) only when *fr* >15 Hz

[a] It is unusual to use Paw greater than 25 cm H_2O, although it may be necessary during lung volume recruit'ent maneuver. Always use a high-volume strategy after lung recruitment; for details of the lung recruitment scheme, see text.

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In an ex vivo lung model, decreases in ΔP demonstrated the recruitment of the lung after increments in Paw. After reaching the recruitment of the lung, ΔP did not decrease more. This also was demonstrated in an in vivo animal lung study, where the decreases in ΔP also correlated with increases in Pa_{O_2} until the recruitment of the lung was achieved; then, no further increases in Pa_{O_2} or decrease in ΔP were observed. So, the changes in ΔP, linked to a progressive increase in Paw, can be used to identify lung recruitment even earlier than oxygenation improvement.[64]

BEST PRACTICES BOX

- Prevention of invasive mechanical ventilation and help to establish the functional residual capacity at delivery is the best respiratory support of immature infants at risk of developing respiratory failure.
- If invasive mechanical ventilation is needed, then prevent lung trauma using the lower tidal volume possible during ventilation.
- If a low compliance lung needs high respiratory pressures to maintain ventilation, the use of HFOV as a rescue therapy can be an effective and safe alternative to conventional ventilation.
- Monitoring and controlling of the tidal volume during HFOV improves its clinical use.
- Using the lower tidal volume at the higher respiratory rate in HFOV can reduce lung injury.

CLINICS CARE POINTS

- Immature infants are at risk of lung injury and aggressive invasive ventilation must be prevented.
- Ventilator lung injury is mostly related to the use of large tidal volumes and can trigger chronic respiratory failure in premature infants.
- HFOV uses low tidal volumes and in several studies has demonstrated a protective effect compared with conventional ventilation.
- The use of HFV is effective as a rescue therapy in severe RDS in premature infants and it is safe as long term studies have demonstrated.
- Accurate measurement of the tidal volume and control of this volume by the use of VG combined to HFOV can be a new protective ventilator strategy to prevent lung trauma in the more immature infants with respiratory failure.

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